

CPSC 317: Introduction to Computer Networking

Module 0: Introduction and Internet Design

Things to do today

- Introduce ourselves
- Talk a little about the course content
- Talk a little more about the course structure and rules

Who am I?

- Ibtissem Bouacheria
 - Sessional Lecturer, Department of Computer Science
 - 5+ years teaching experience
- Office: ICCS 249
- My interests: networking, Internet Of Things, routing protocols

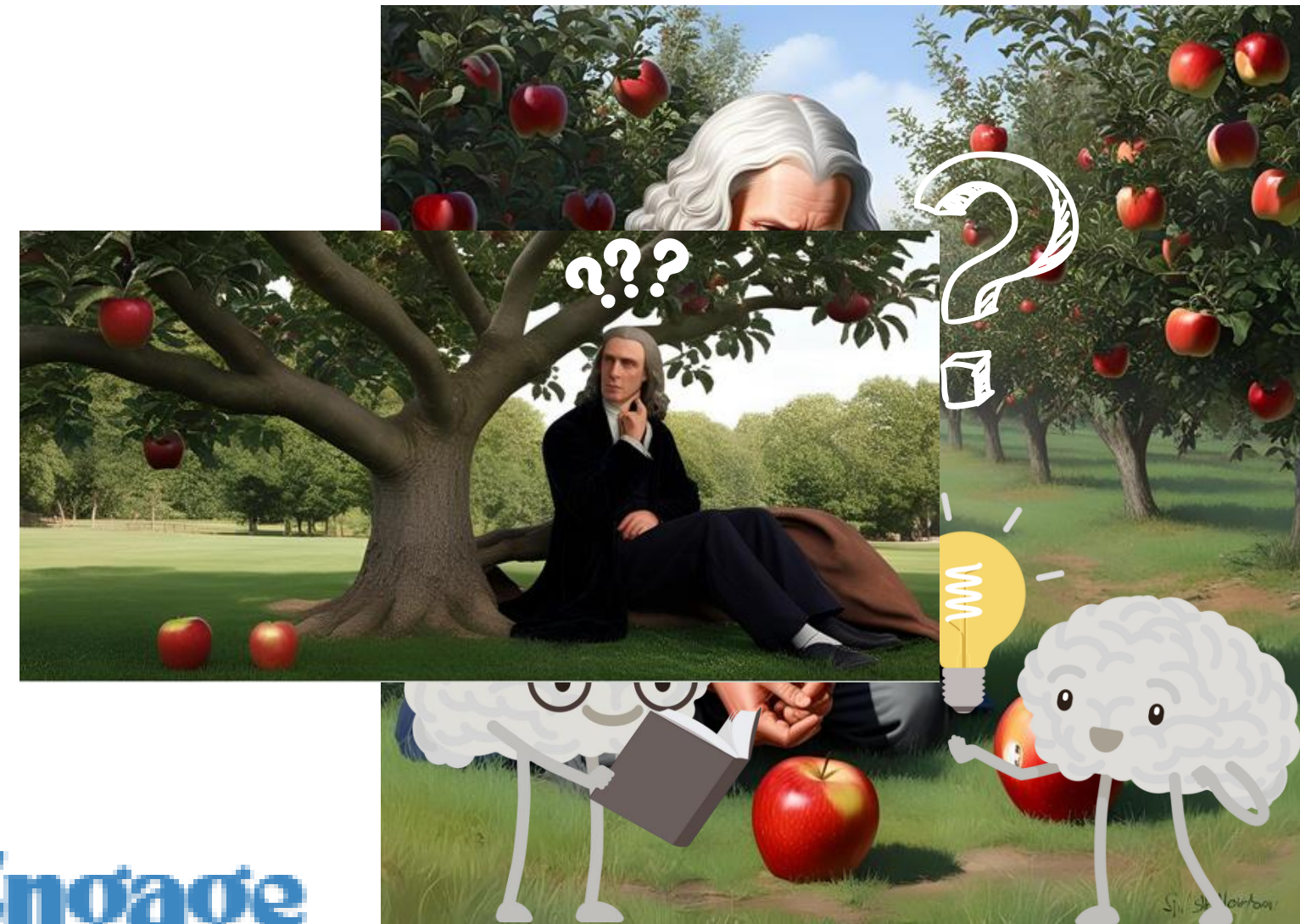
Who am I?

Quote that shaped my life

“Knowledge can be taken, but not given”

What do you think?

“Knowledge can be taken, but not given”



Stay Curious ! Ask Questions ? Fully Engage & Learn

Introduce Yourself

- Talk to your neighbours
- Introduce yourself to each other

Topics of discussion

- What area of Computer Science are you interested in?
- Why are you taking this course?
 - What do you want to learn?

Pre-Requisites

- CPSC 213 (Introduction to Computer Systems)
- CPSC 221 (Basic Algorithms and Data Structures)
- If you don't have these prerequisites, check
 - <https://www.cs.ubc.ca/prerequisite-rules>

Course Information

❖Lecture:

- Instructor: Ibtissem Bouacheria
- Section 911
 - Class time : Tue/Thu at 11:00AM – 2:30PM

Course Information

❖ **Tutorials:** Sections D1A/D1B/D1C/D1D

–Tutorial time :

- D1A: Tue/Thu at 03:30PM – 04:30PM
- D1B: Tue/Thu at 04:30PM – 05:30PM
- D1C: Tue/Thu at 05:30PM – 06:30PM
- D1D: Tue/Thu at 06:30PM – 07:30PM

Tutorials

- **What will be covered in tutorials?**
 - Practical exercises based on course content and programming assignments.
- **How many tutorials are there?**
 - 10 tutorials in total.
- **When do tutorials start?**
 - They begin this week.

Tutorial Registration

- **Do I have to go to a tutorial?**
 - Yes
- **I want a tutorial, but sections don't fit my schedule**
 - Keep looking – there is often quite a bit of “churn” in the first weeks
 - We'll keep an eye on registration numbers
 - Attend other tutorials in the meantime

Course Tools



Canvas

Announcements

Syllabus

Textbook

Additional references

Links to all other tools



Canvas

Announcements

Syllabus

Textbook

Additional references

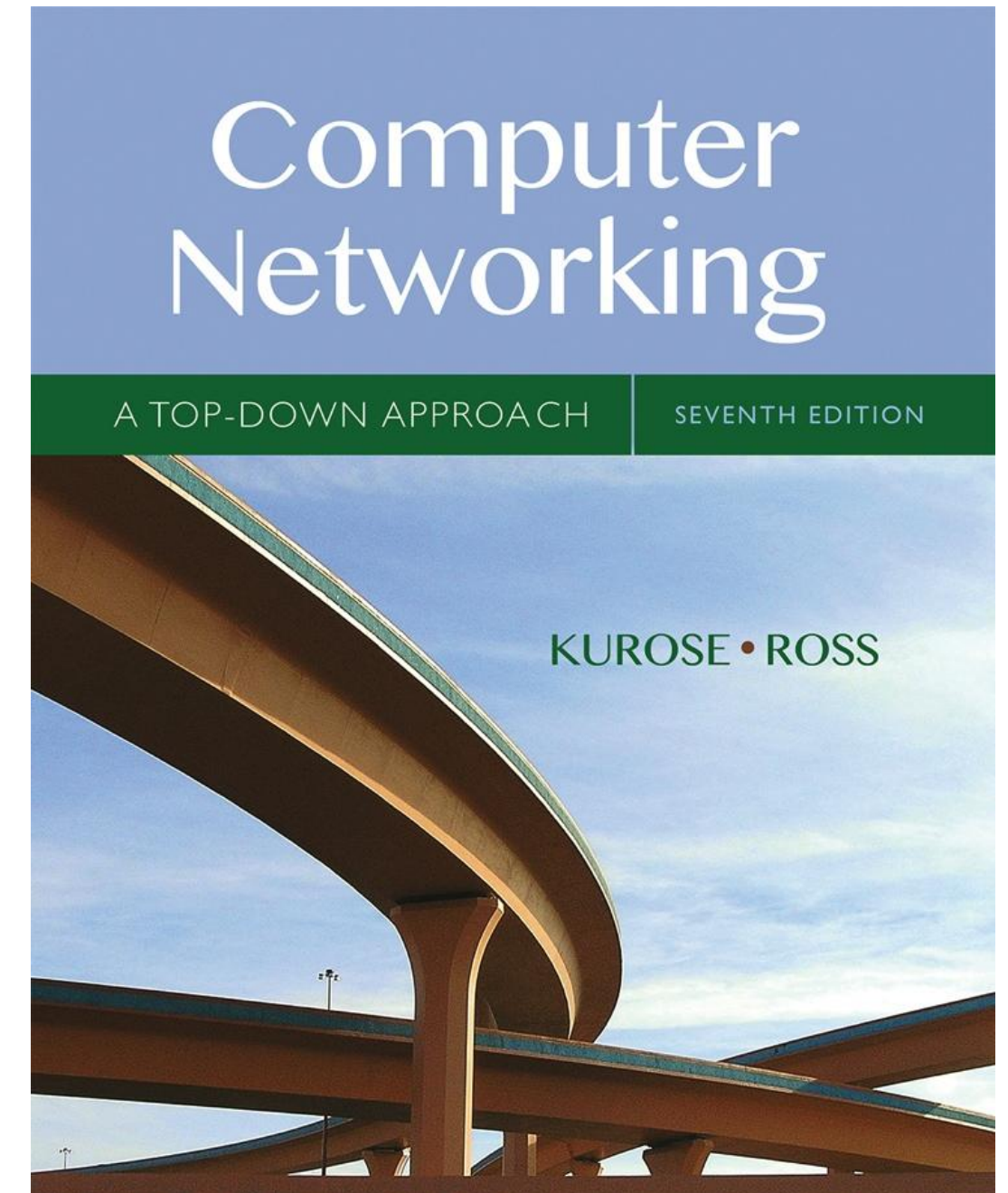
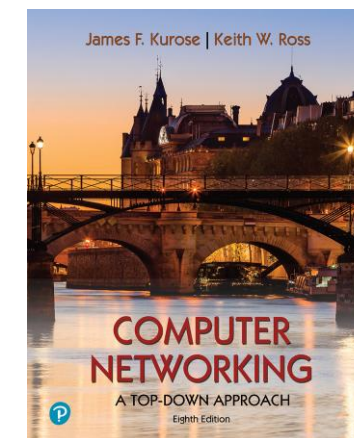
Links to all other tools

Textbook

Computer Networking: A Top-Down Approach,

7th edition,
Jim Kurose, Keith Ross,
Pearson, 2017

Use 8th edition if you're planning
to take CPSC 417 in the future



Course Tools



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iClicker

In-class

Participation

Questions

(3% Bonus Grade)



In-class
Participation
Questions

(3% Bonus Grade)

iClicker Cloud

- In-class participation questions
 - Typically, multiple-choice or short answer questions
- 3% Bonus Grade
- Answering gets you half the points per question
 - Answering correctly gets you the other half
- Please use a valid student ID when logging in
- Section 911: Join: <https://join.iclicker.com/MRYF>

Course Tools



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Textbook

Additional references

Links to all other tools

(**3%** Bonus Grade)



iClicker

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Questions

PrairieLearn

PrairieLearn

Practice Part

In-Class Activities (ICA)

Quizzes (PFQ)

Graded Part

Assignments (**35%**)

Quizzes (**30%**)

Final Exam (**35%**)

Request a review

PrairieLearn

PrairieLearn

Practice Part

In-Class Activities (ICA)
Quizzes (PFQ)

Graded Part

Assignments (35%)

Quizzes (30%)
Final Exam (35%)

Request a review

- **Programming Assignments** (35% of your final grade)
 - Will be released and **submitted on PrairieLearn**
 - You will have **five assignments**
 - PA1: Implementing a DICT client
 - PA2: Implementing a DNS client
 - PA4: Implementing a reliable transport protocol
 - PA5: Implementing forwarding in a router
 - **Submissions** may have **preliminary autograding on PrairieLearn**
 - **TAs will still review** all submissions
 - All **assignments** to be done **individually**, except PA4

PrairieLearn

PrairieLearn

Practice Part

In-Class Activities (ICA)
Quizzes (PFQ)

Graded Part

Assignments (35%)

Quizzes (30%)

Final Exam (35%)

Request a review

Quizzes and Final Exam Format

- Short answer questions / Multiple-choice questions
- Computer-Based Testing Facility (CBTF)
- Synchronous
- **Quizzes** (30% of your final grade)
 - Weekly quizzes
 - In CBTF, throughout the week
 - First quiz self-scheduled from May 20th to May 23rd.

Final Exam (35% of your final grade)

- Do not schedule travel in April until you know when the exam is happening (and even then, be conservative – unforeseen events (fire alarms, etc.) can cause the final exam schedule to change at the last minute)

Course Tools



Canvas

Announcements

Syllabus

Textbook

Additional references

Links to all other tools

(3% Bonus Grade)



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PrairieLearn

Practice Part

In-Class Activities (ICA)

Quizzes (PFQ)

Graded Part

Assignments (35%)

Quizzes (30%)

Final Exam (35%)

Request a review



GitHub

Slides



Slides

Slides

- Slides will be regularly posted on GitHub and Canvas
- As much as possible will be posted in advance
 - Sometimes they may be modified shortly before class

<https://github.students.cs.ubc.ca/cpsc317/Slides-2026S1>

Course Tools



Canvas

Announcements

Syllabus

Textbook

Lecture Slides

Additional references

Links to all other tools (3% Bonus Grade)



iClicker

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PrairieLearn

Practice Part

In-Class Activities (ICA)

Quizzes (PFQ)

Graded Part

Assignments (35%)

Quizzes (30%)

Final Exam (35%)

Request a review



GitHub

Slides



Piazza

Discussion board



Piazza

Discussion board

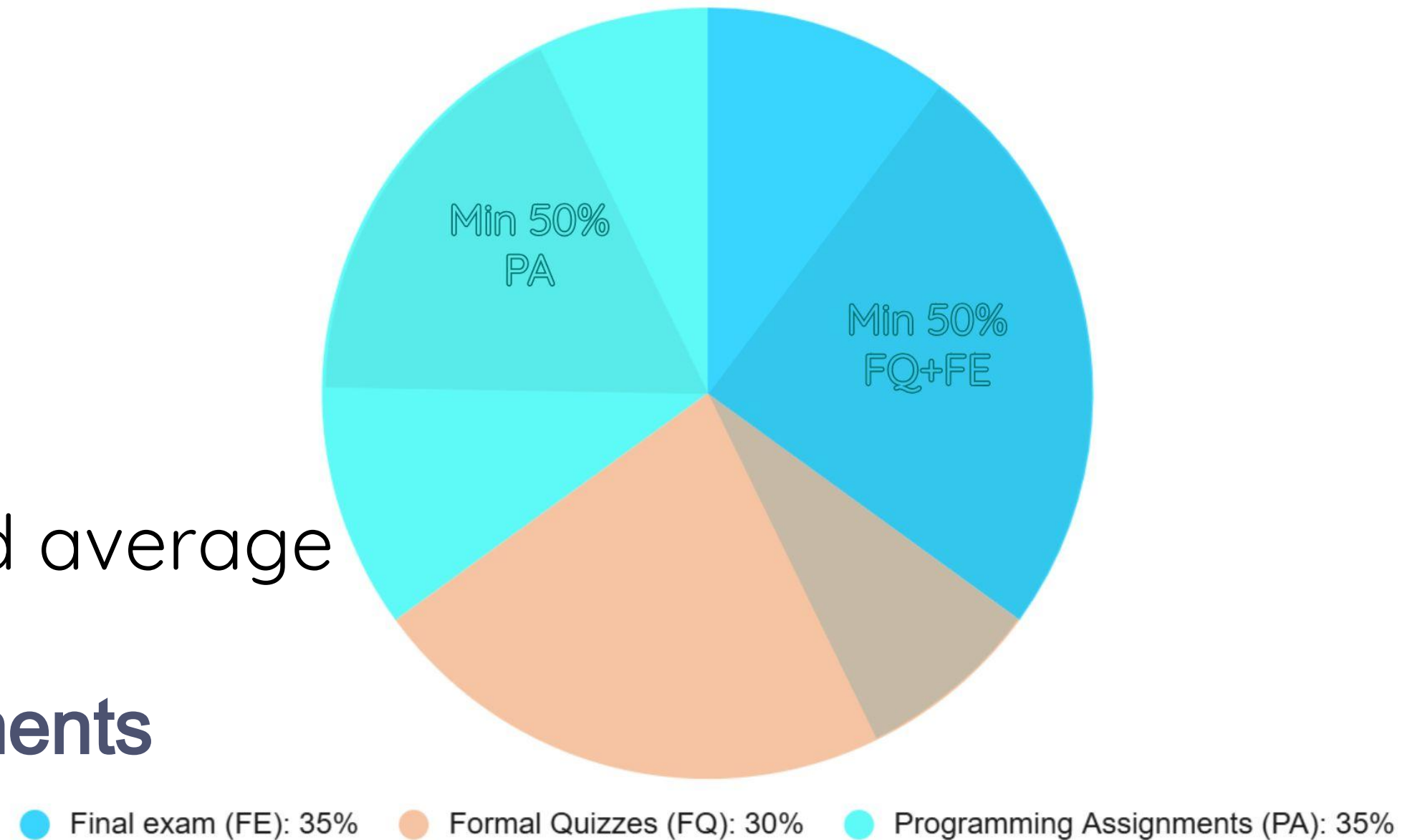
Piazza

- Discussion board
 - Do not use the discussion board on Canvas
- To join:
 - Use the link below. It is provided on Canvas (or search Piazza for this course)

https://piazza.com/ubc.ca/summer2026/cpsc_v3179112026ss

Grading Scheme

- **Programming Assignments: 35%**
- **Quizzes: 30%**
- **Final Exam: 35%**
- You **must** get **50%** in the weighted average of the **quizzes** and the **final exam**.
- You **must** get **50%** on the **assignments**
- **Participation:**
 - Lectures: up to **3% Bonus** (based on active presence and iClicker questions in class)
 - Tutorials: up to **2% Bonus** (based on active presence)
- We reserve the right to make minor modifications to this scheme



Contact Information

- For **most requests email**: cpsc317-admin@cs.ubc.ca
 - Extensions, Illness accommodations, ...
- To **contact the instructor only**: ibtissem.bouacheria@ubc.ca
- **Our teaching assistants** are:
 - Han, Kevin, Kyle and Satvik
- **Office hours**: To be announced on Piazza
- For **general questions** about **course material or assignments**, **do not email** directly. Instead:
 - Use **Piazza**
 - Ask a TA during **office hours**
 - Ask instructor **before or after class** or during office hours

Some Rules: Absences

- If you must be absent in a quiz, contact the course coordinator coordinator (cpsc317-admin@cs.ubc.ca) as soon as you are aware of the problem
- You do not need to notify us of absences in days when there is no graded work
- For assignments, there is no special consideration for expected absences
- Full policy and notification instructions available on course syllabus page.

Some Rules: Marking Disagreements

- You have 1 week from when a piece of graded work is handed back to request a review
 - Review requests must be submitted through PrairieLearn as reported issues
 - After 1 week the mark stands
- We may review marking of other questions
- Full policy and notification instructions available on Course Syllabus

Academic Conduct

- Don't cheat!!
- Review University and course policies on academic misconduct, as well as any relevant assessment instructions
- If you are uncertain: ASK!!!
- We can and will compare your solutions to those of other students (past and present)
- Possible penalties
 - Failing grade in the work in question or in the course in general
 - Suspension from the University
 - Reprimand with letter in student's file
 - Notation in student's permanent record
- Full policy and information on course syllabus

Things To Do

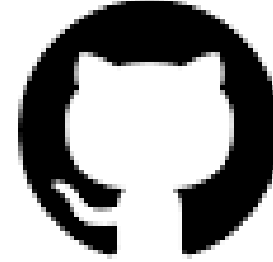
- Get familiar with course tools (if you are not already):



Canvas



iClicker



GitHub

PrairieLearn

PrairieLearn



Piazza

<https://canvas.ubc.ca/courses/186807>

<https://join.iclicker.com/MRYF>

<https://github.students.cs.ubc.ca/cpsc317/Slides-2026S1>

https://us.prairielearn.com/pl/course_instance/215049

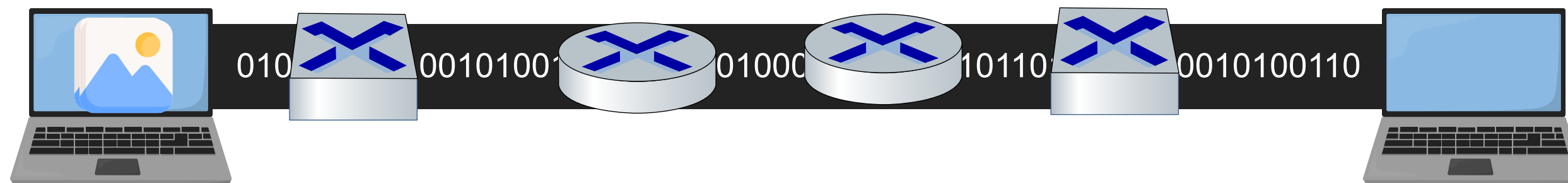
https://piazza.com/ubc.ca/summer2026/cpsc_v3179112026ss

Reflexion

- How do we make it work?
 - Communication media (wires, radio signals, etc.)
 - Network architecture (hosts, NICs, switches, routers, etc.)
 - Protocols
 - Applications

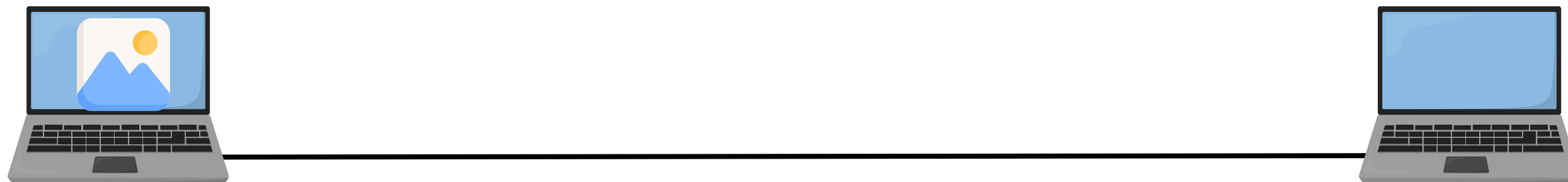


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Reflexion

- **What is a computer network used for?**
 - Communication
 - Information sharing
 - Task distribution
 - Resource sharing
 - Scaling and redundancy



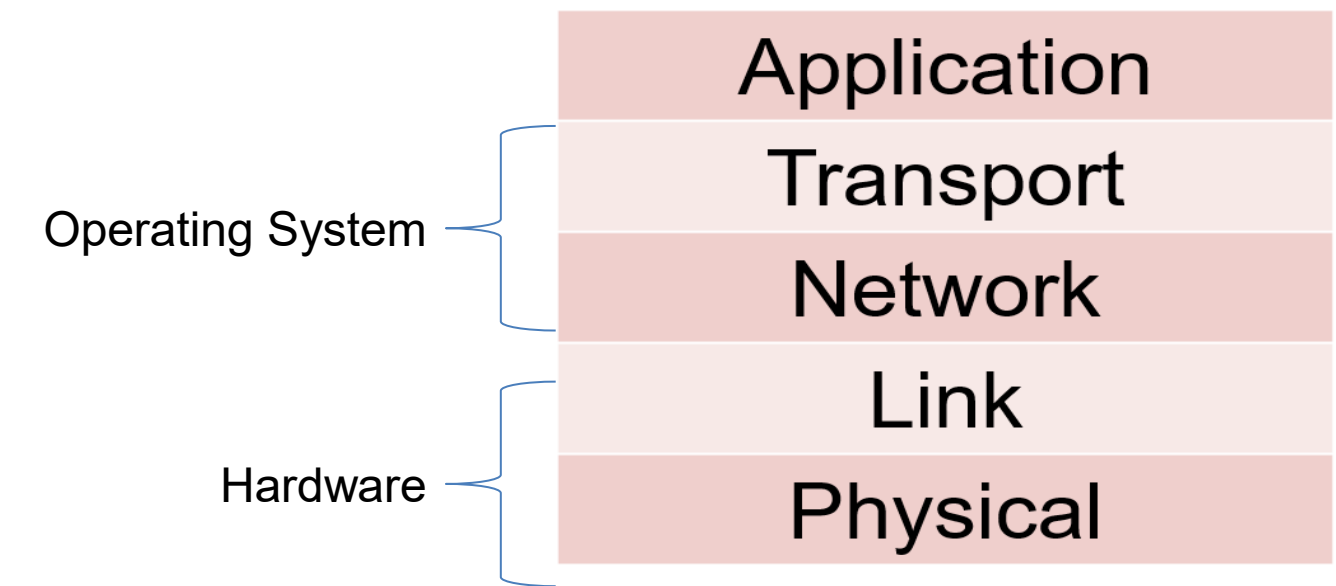
Coping with complexity

- **Complexity** limits the size of the systems we build
 - We need techniques to cope with it in order to build larger systems
- **Modularity**: Divide system into smaller components
- **Abstraction**: Define how a component interacts with the outside world. Then don't worry about it works internally



Coping with complexity (cont.)

- **Robustness drives independence**
 - Be tolerant of inputs (deal with incorrect values sensibly)
 - Be strict on outputs (don't produce unexpected values)
- **Layering**
 - A particularly successful type of modularity
 - The system contains a number of levels, each built on the previous one
 - Used in networks and operating systems
- **Hierarchy**
 - A more general approach when layering is too strict
- **Indirection**
 - Names are the glue that holds systems together
 - Using indirection allows sub-systems to be swapped more easily



Why Study Computer Networking?

- Unlike most courses the focus is on **COMMUNICATION** not computation.
- What is different? Communication requires at least two parties.
- Plumbing -- what underlies the applications in wide use today.
- The applications enabled by networking are the most disruptive technology over the past 30 years

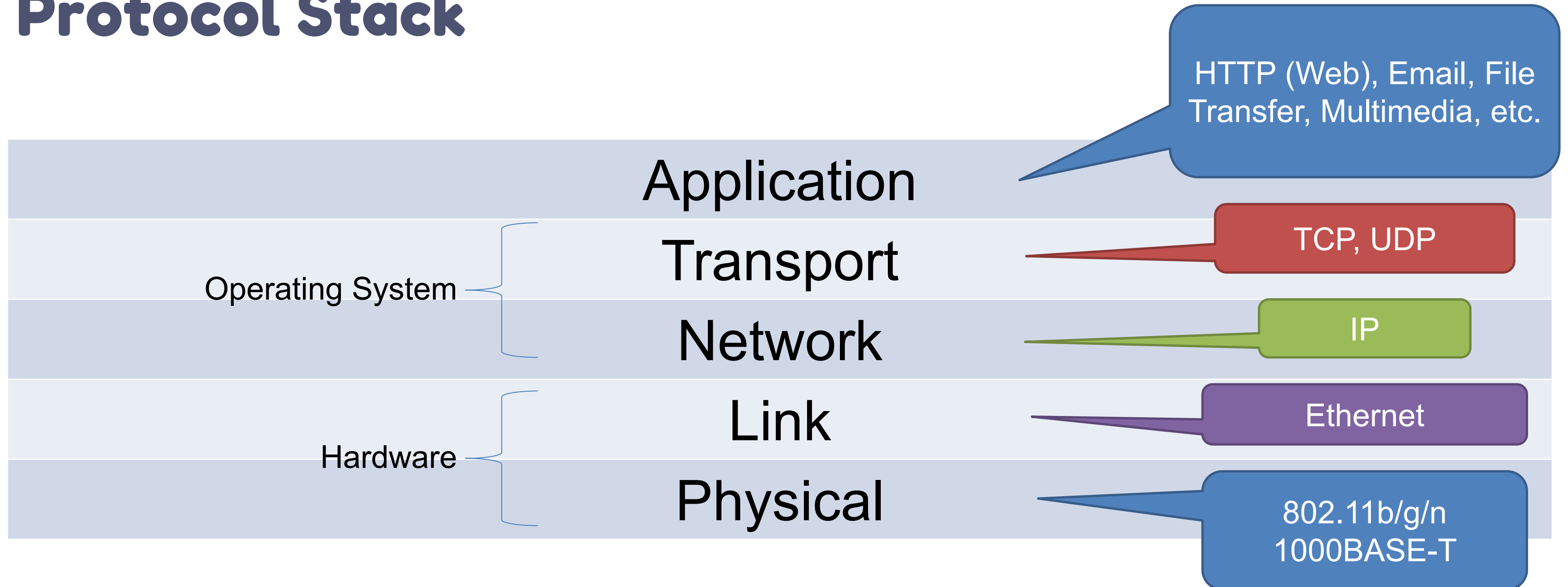
Course Objectives

- You should become comfortable with:
 - writing and working with different types of programs that use computer networks
 - terminology about networking

Course Objectives

- Key paradigms and strategies used in developing applications that use networks:
 - Isolation and privacy
 - Data loss
 - Performance
 - Naming and location
 - Using layers and abstractions
- Applying these strategies and paradigms
- Basic concepts on how the Internet is put together and works

Protocol Stack



Next Topic: Design of the Internet